

A Proposal of Mach-Zehnder Interferometer

RYUNOSUKE SAITO

ryunosuke.saito@advantest.com

1. Introduction

The Mach-Zehnder Interferometer (MZI) is one of the most fundamental components in silicon photonics. By splitting an optical signal into two paths and recombining it after a controlled phase shift, the MZI enables precise manipulation of light through interference. Owing to its simple structure, MZIs are widely used for photonic integrated circuits. Typical applications include optical switches, wavelength filters, and wavelength demultiplexers (DEMUXs). In this report, simulation results of MZIs are presented and analyzed.

2. Theory

Fig. 1. shows an unbalanced MZI. MZI consists of a Y-branch splitter, strip waveguides, and a Y-branch combiner. In this report, the phase difference between the upper and lower arms is assumed to be generated by a path length difference rather than by a difference in effective refractive index (n_{eff}).

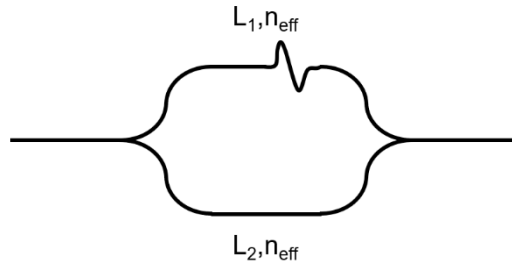


Fig. 1. Unbalanced MZI schematic illustration

First, the electric field of input port (E_i) is split to upper and lower paths. Electric field of upper path (E_1) and lower path (E_2) are given by

$$E_1 = E_2 = \frac{E_i}{\sqrt{2}} \quad (1)$$

Under the assumption that the upper and lower waveguides are lossless and possess the same n_{eff} , the electric fields at the inputs of the Y-branch combiner, E_{o1} and E_{o2} , are given by

$$E_{o1} = E_1 e^{-i\beta L_1} \quad (2)$$

$$E_{o2} = E_2 e^{-i\beta L_2} \quad (3)$$

β is the propagation constant of the waveguide. It is given by $\beta = \frac{2\pi n_{eff}}{\lambda}$. The electric fields are combined by the Y-branch combiner, resulting in the following output field (E_o):

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \frac{E_i}{2} (e^{-i\beta L_1} + e^{-i\beta L_2}) \quad (4)$$

Since the optical intensity is proportional to the square of the electric field magnitude, the output intensity (I_o) is calculated as

$$\begin{aligned}
I_o &= \frac{I_i}{4} |e^{-i\beta L_1} + e^{-i\beta L_2}|^2 \\
&= \frac{I_i}{2} [1 + \cos(\beta \Delta L)]
\end{aligned} \tag{5}$$

For an unbalanced MZI, the free spectral range (FSR) is known to be given by

$$FSR(\lambda) = \frac{\lambda^2}{\Delta L n_g} \tag{6}$$

where n_g is the group index of the waveguide. It should be noted that this expression is derived under the assumption that the interferometer operates at a high interference order.

3. Modeling and Simulation

3.1 Waveguide simulation

I want to design two strip waveguides. One is 220nm height and 500nm width. The other is 220nm height and 350nm width. The waveguide widths were chosen such that the waveguides operate in the single-mode regime and support propagation of the fundamental quasi-TE mode. Fig.2. shows the electric fields intensity of quasi-TE mode. Fig.3. shows the effective index simulated from numerical MODE solution. It can be seen that the 220 nm × 350 nm waveguide exhibits a lower effective refractive index because a larger portion of the optical mode extends into the cladding.

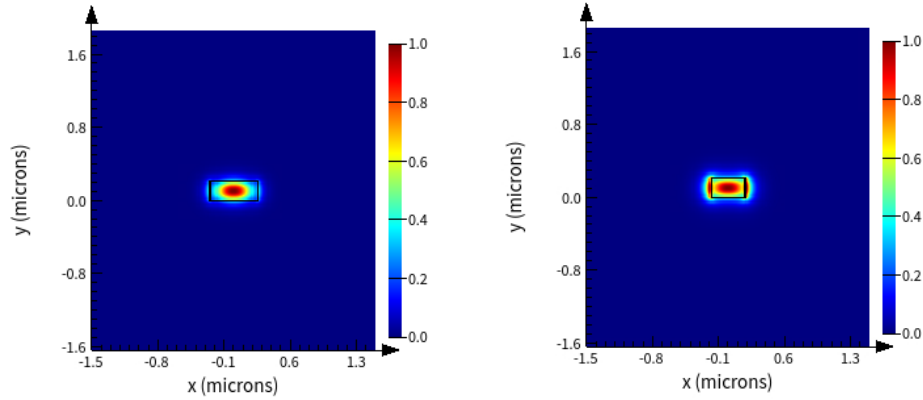


Fig. 2. Electric fields intensity of TE mode (Left: 500nm×220nm Right: 350nm×220nm).

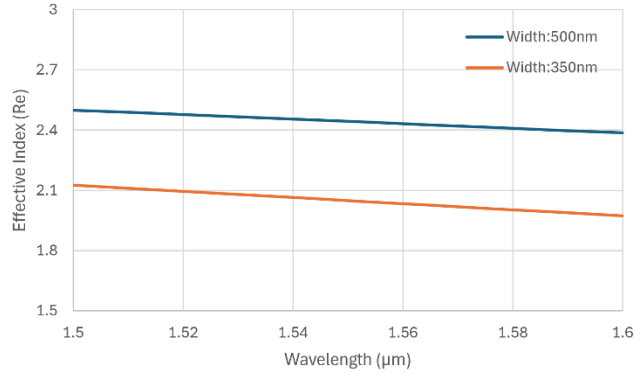


Fig. 3. Effective index simulated from Lumerical MODE solution

Fig.4. shows group index. Since the wavelength dependence of the group index differs between the two waveguides, their impact on MZI characteristics can also be compared.

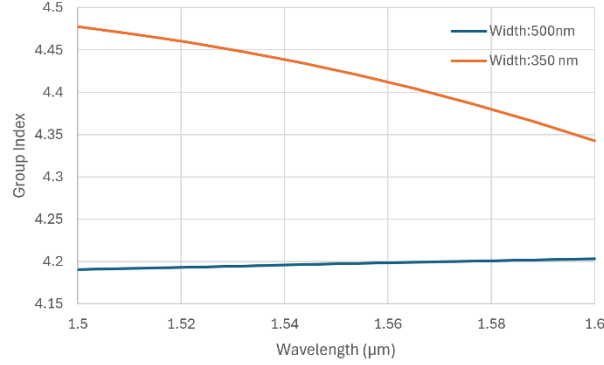


Fig. 4. Group index simulated from Lumerical MODE solution

3.2 Waveguide modeling

A polynomial fitting model was developed based on the simulation results. The resulting polynomial model is given as follows:

$$\text{Width:350nm} \quad n_{eff}(\lambda) = 2.049 - 1.531(\lambda - 1.55) + 0.43(\lambda - 1.55)^2$$

$$\text{Width:500nm} \quad n_{eff}(\lambda) = 2.443 - 1.132(\lambda - 1.55) - 0.04(\lambda - 1.55)^2$$

Fig.5. shows the simulated effective refractive index together with the developed waveguide fitting model.

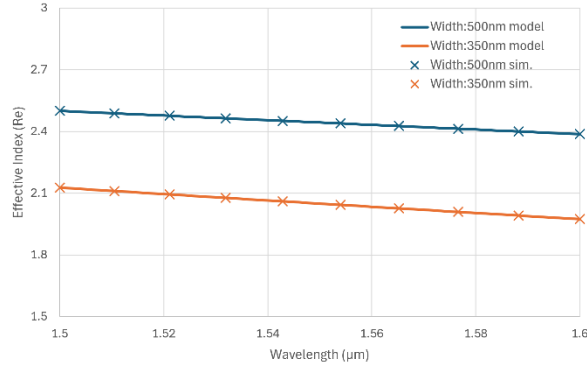


Fig. 5. Waveguide polynomial fitting model.

3.3 MZI simulation

Fig.6. shows MZI simulation system in Lumerical INTERCONNECT. The grating couplers and Y-branches used in this work were obtained from the PDK. Simulations were performed using the waveguide models derived from the simulation results. Table 1 summarizes the design parameters of the MZI considered in this study and the FSR values predicted from Eq. (6). Fig.7. shows the transmission spectrum of the MZI with a waveguide width of 500 nm, while Fig.8. shows the transmission spectrum of the MZI with a waveguide width of 350 nm. The FSR was extracted from these simulation results and compared with the values predicted by Eq. (6). The comparison results are presented in Figure 9. It can be observed that the deviation between the simulated FSR and theoretical value increases as ΔL decreases. This trend is likely

because the assumption used in deriving Eq. (6), namely that the interferometer operates at a high interference order, is no longer valid for small ΔL .

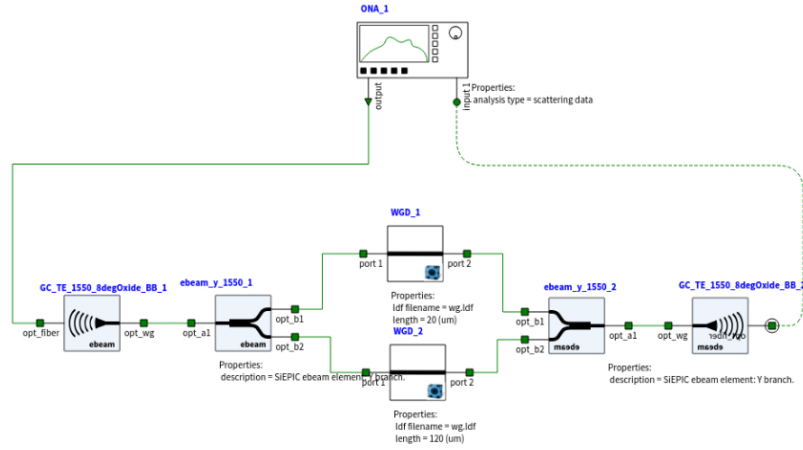


Fig. 6. MZI simulation system.

Table 1. MZI design parameters

Waveguide height [nm]	Waveguide width [nm]	ΔL [um]	FSR [nm]
220	500	0	-
220	500	20	28.601
220	500	50	11.441
220	500	100	5.720
220	350	0	-
220	350	20	27.178
220	350	50	10.871
220	350	100	5.436

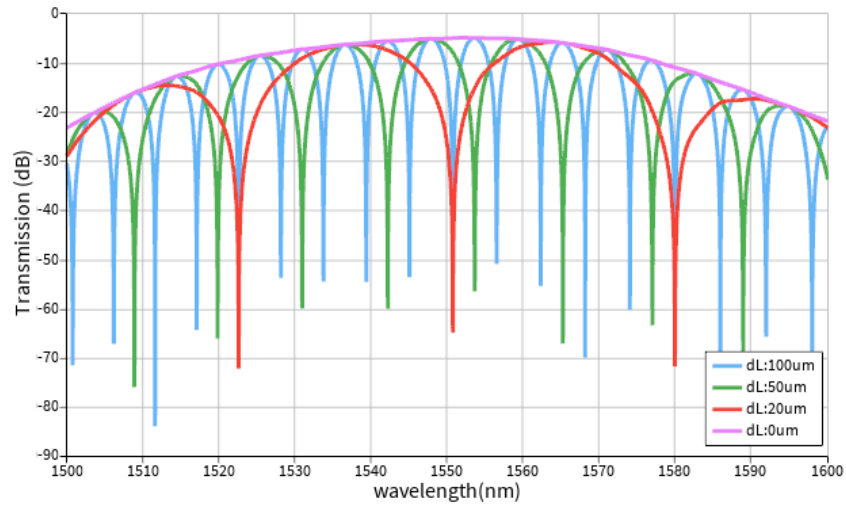


Fig.7. Transmission spectrum of the MZI with a waveguide width of 500nm

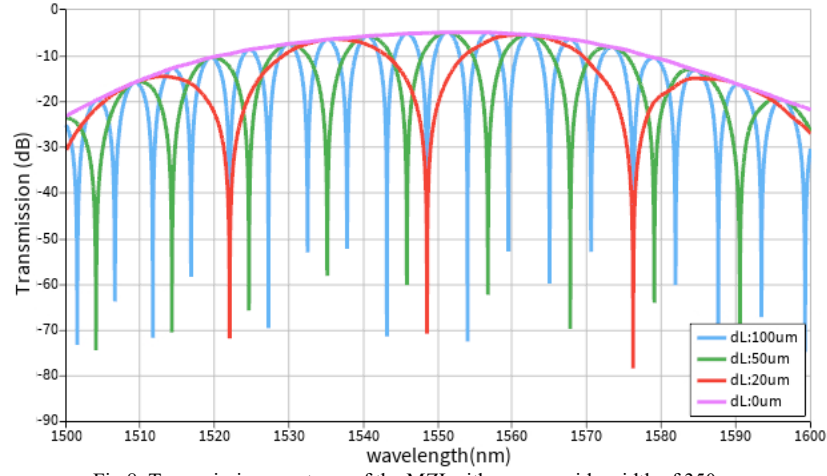


Fig.8. Transmission spectrum of the MZI with a waveguide width of 350nm

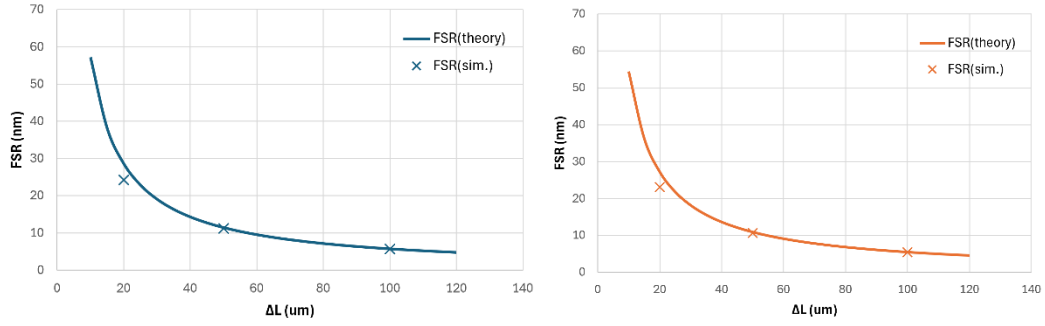


Fig.9. Comparison between theoretical and simulated FSR values

4. Conclusion

In this report, the interference theory of an unbalanced MZI was first presented. Compact models of the waveguides were then developed from mode simulation results and employed in MZI circuit simulations. The transmission spectra of the designed MZIs were analyzed, and the resulting FSR values were compared with theoretical predictions. The comparison revealed the level of agreement between the analytical model and the simulation results.

Based on the simulation results presented in this report, the MZI designs summarized in Table 1 are proposed for fabrication and experimental evaluation.